B10.7 Device Assignment (DA) and Coherent Device Assignment (CDA)

The term, RME-DA, refers to a device that is IO coherent.

The term, RME-CDA, refers to a device that has the capability of being fully coherent.

> **Note**
>
> The expected use for RME-DA and RME-CDA in conjunction with CHI is using chip(let)-to-chip(let) links. For more information on chip-to-chip connections in CHI, see *AMBA® CHI Chip-to-Chip (C2C) Architecture Specification*.

B10.7.1 Introduction

RME-DA and RME-CDA enables the secure assignment of assignable device interfaces.

A *Device Permission Table* (DPT) contains the permission attributes associated with physical addresses and specifies a corresponding set of DPT checks that apply to incoming accesses from devices.

Figure B10.1 shows the support for RME-DA and RME-CDA in multiple accelerator device topologies, where the accelerators are connected to a host.

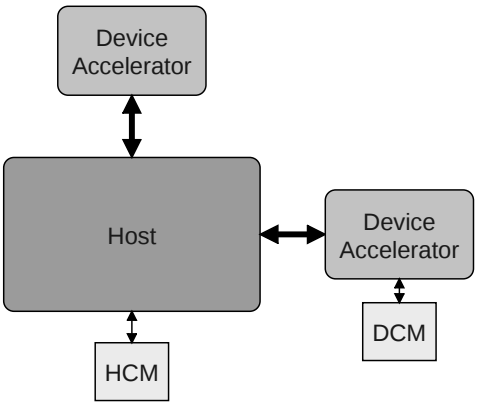

Figure B10.1: Host connected to two Accelerator devices

A host, made from a single or multiple SoC dies, forms a *Trusted Computing Base* (TCB).

A host:

- Is required to enforce the security policies required by the system.
- Can be connected to multiple devices.
- Can have memory termed as *Host Coherent Memory* (HCM) directly attached to it. The HCM can be coherently accessed by the host, or by any fully coherent or IO coherent devices.
- Must not send a Stash Snoop to a device, unless the line is already legally permitted to be cached by the device.
- Is permitted to send DVMOp requests or SnpDVMOp snoops towards a device.

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A device is a single physical chip(let) and is treated as a single unit from a trust perspective and individually attested.

A device:

- Is trusted in its entirety for a set of Realms only and trusted to maintain Realm separation where required.
- Can have private memory-mapped resources.
- Can have one or more interfaces.
- Can have memory termed as *Device Coherent Memory* (DCM) directly attached to it. The DCM can be coherently accessed by the host, the device, and by other coherent devices.
- Can access the DCM of another coherent device only where that access is channeled through a host compute node.
- Only operates in the Realm and Non-secure PAS.
- Is expected, but not required, to include the following fields in all requests to the host:
  - SecSID1
  - StreamID
- Only receives snoop requests for:
  - Regions of memory owned by Realms that trust the device
  - Non-secure regions of memory
- Cannot be directly connected to another device.
- Is not permitted to send DVMOp requests or SnpDVMOp snoops towards to a host.

The host and the devices can have multiple Request Nodes and Home Nodes.

It is expected that the host will perform additional checks on certain messages from or to a device. These checks ensure that:

- Requests coming from the device to HCM are permitted to access that memory. This means the device cannot issue a request with a random PA and expect to get access to that location.
- Snoops coming from the device are only for DCM. This means the device cannot issue a request with arbitrary PA to gain access to a location.
- Snoops going to the device for cached HCM locations only expose accesses that the device is permitted to observe. This means the device cannot build up a picture of all trusted system activity for more targeted attacks.

For more information on RME-DA and RME-CDA, see *Arm® Architecture Reference Manual for A-profile architecture*.

B10.7.2 Field applicability

Support for RME-DA or RME-CDA at an interface is determined by the DevAssign_Support property. The property defines if the SecSID1 and StreamID fields are present on that interface.

The applicability of StreamID and SecSID1 fields in relation to the various messages is detailed in this section:

- B10.7.2.1 *Device request to HCM location or DCM location of a peer device*
- B10.7.2.2 *Host request to DCM location*
- B10.7.2.3 *Device snoop to cached DCM location*
- B10.7.2.4 *Host snoop to cached HCM location or cached DCM location of a peer device*

The omission of a field name indicates that the related field is not applicable in the associated message.

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B10.7.2.1 Device request to HCM location or DCM location of a peer device

Figure B10.3 shows the fields that are applicable for device requests to HCM memory locations, or to DCM location of a peer device.

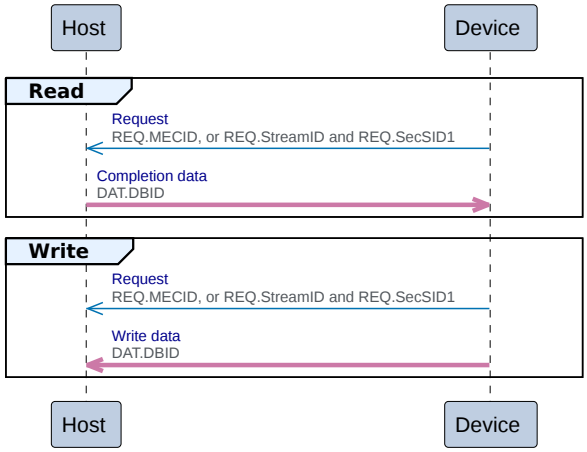

Figure B10.2: RME-CDA fields in device requests to DCM locations

When DevAssign_Support is Device_StreamID_SecSID1, the common field on the REQ channel with StreamID and MECID is considered as StreamID, regardless of the MEC_Support value.

When the DevAssign_Support property is not Device_StreamID_SecSID1:

When the MEC_Support property is True
The common field on the REQ channel with StreamID and MECID is considered MECID.

When the MEC_Support property is False
Neither StreamID or MECID are applicable.

On the DAT channel, the common field shared by DBID and MECID is considered DBID.

B10.7.2.2 Host request to DCM location

Figure B10.3 shows the fields that are applicable for host requests to DCM memory locations.

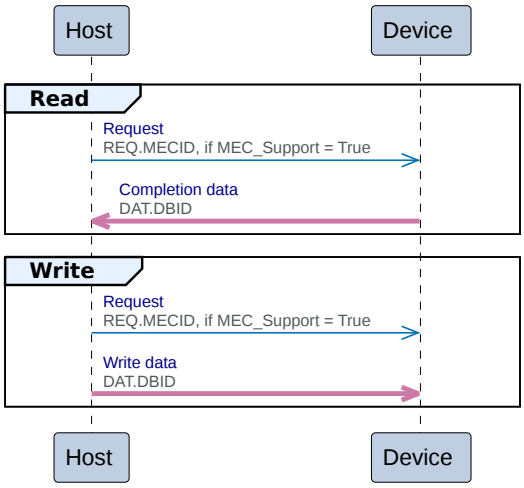

Figure B10.3: RME-CDA fields in host requests to DCM locations

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On the DAT channel, the common field shared by DBID and MECID is considered DBID.

B10.7.2.3 Device snoop to cached DCM location

Figure B10.3 shows the fields that are applicable for device snoops to DCM memory locations that have been cached in the host.

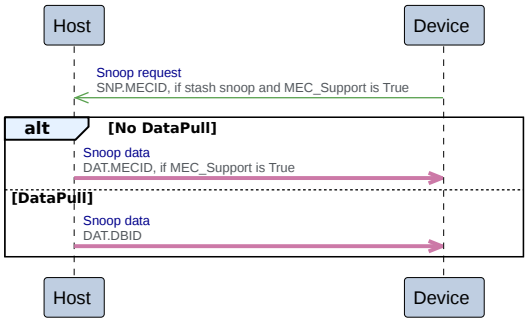

Figure B10.4: RME-CDA fields in device snoop to cached DCM locations

For a snoop response with data that includes a DataPull request, the common field shared by DBID and MECID is considered DBID.

For a snoop response with data that does not include a DataPull request, if MEC_Support is True, the common field on the DAT channel shared by DBID and MECID is considered MECID.

B10.7.2.4 Host snoop to cached HCM location or cached DCM location of a peer device

Figure B10.5 shows the fields that have been cached in the device and are applicable for host snoops to HCM memory locations, or DCM locations of a peer device.

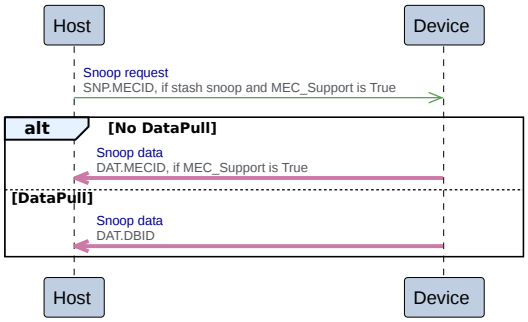

Figure B10.5: RME-CDA fields in host snoop to cached HCM locations

For a snoop response with data that includes a DataPull request, the common field shared by DBID and MECID is considered DBID.

If MEC_Support is True, for a snoop response with data that does not include a DataPull request, the common field on the DAT channel shared by DBID and MECID is considered MECID.

B10.7.3 Component requirements

This section describes the host and device requirements.

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B10.7.3.1 Host requirements

Table B10.1 shows the host requirements depending on the MEC_Support and DevAssign_Support property values.

Table B10.1: Host requirements

Scenario	MEC_Support	Description
Device request to a HCM location or DCM location of a peer device	False	The host performs the following checks: – A DPT check might be required to check the Security state, indicated by SecSID1, has granted the StreamID with permission to access the physical address. – A GPC is required to verify that the request PA and PAS combination is permitted. If these checks pass, the request proceeds normally. If these checks fail, the host must respond in a protocol-compliant manner. The responses are marked as NDERR.
	True	All of the False requirements apply, and: **When PAS is Realm**: The MECID value can be non-zero. **When PAS is Non-secure**: The MECID value must be zero. **When DevAssign_Support = Device_StreamID_SecSID1**: If DPT and GPC checks pass, the MECID value from the *Stream Table Entry* (STE) is used as part of the request as it progresses into the system. **When DevAssign_Support = Device_NoStreamIS_NoSecSID1**: The MECID value provided continues with the request as it progresses into the system.
Host request to DCM location	False	Realm and Non-secure are the only permitted PAS values.
	True	All of the False requirements apply, and: **When PAS is Realm**: The host must populate the MECID field in the request with a valid value. **When PAS is Non-secure**: The MECID value must be zero.
Device snoop to a cached DCM location	False	The host must check if the snoop address falls within the address range of the DCM attached to the device sending the snoop. If the address is found to be: – In the allowed DCM range, the snoop must be processed as normal. – For a response to a stashing snoop where a DataPull is requested, the host must provide a DBID value. – Not in the allowed DCM range, the host must terminate the snoop and return a SnpResp_I response to the device.
	True	All of the False requirements apply, and if the snoop is processed as normal, any associated SnpRespData response must include: – The MECID value, if DataPull is 0. The inclusion of the MECID value in the snoop response enables the device to encrypt the data before writing back to DCM. – The DBID value, if DataPull is 1.

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Table B10.1 – *Continued from previous page*

Scenario	MEC_Support	Description
Host snoop to a cached HCM location or cached DCM location of a peer	False	The following requirements apply: **When PAS is Realm** — The host is permitted to forward the snoop, if it specifies a location that is known to be present in the device cache. A Home can filter snoops using a tracker to follow device caching precisely, such as a Snoop Filter. **When PAS is Non-secure** — No checks are required. The host port must respond with IMPLEMENTATION SPECIFIC data values that are fully poisoned if the host is unable to send a snoop request across the link to the device, and the host snoop filter tracks the location in the device as being in a Dirty state.
	True	All of the False requirements apply. If the host issues a stash snoop to the device: **When PAS is Realm** — The host must populate the MECID field in the request with a valid value. **When PAS is Non-secure** — The MECID value must be zero. In the scenario where the snoop cannot be sent and the host port has to send fully poisoned data, if the host port cannot associate the response with a correct MECID, it must use the default MECID value of zero.

B10.7.3.2 Device requirements

Table B10.2 shows the device requirements depending on the MEC_Support and DevAssign_Support property values.

Table B10.2: Device requirements

Scenario	MEC_Support	Description
Host request to DCM location	False	No additional requirements.
	True	The device must use the received MECID value for accessing DCM memory.
Device request to a HCM location or DCM location of a peer device	False	The following requirements apply: **When DevAssign_Support = Device_StreamID_SecSID1** — The device must populate the StreamID and SecSID1 values, when applicable. See Table B10.3 for permitted values and relationship between SecSID1 and PAS. **Note** StreamID corresponds to the PCIe *Requester ID* (RID). SecSID1 corresponds to the PCIe T bit in TLP. **When T = 0** — Access is Non-secure PAS. **When T = 1** — Access is either Non-secure or Realm PAS. **When DevAssign_Support != Device_StreamID_SecSID1** — The StreamID, MECID, and SecSID1 fields are inapplicable and must be driven to zero.

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Table B10.2 – *Continued from previous page*

Scenario	MEC_Support	Description
	True	The following requirements apply: **When DevAssign_Support = Device_StreamID_SecSID1** — The device must populate the StreamID and SecSID1 field values, when applicable. **When DevAssign_Support != Device_StreamID_SecSID1** — The device must populate the MECID field with a valid value. The MECID value can be non-zero when the PAS is Realm. The MECID value must be zero when PAS is Non-secure.
Device snoop to a cached DCM location	False	A device is expected to send snoops with an address within the DCM address range only. Address range checks in the snoops to the host are also carried out on the host side.
	True	All of the False requirements apply, and if the device issues a stash snoop to the host: **When PAS is Realm** — The device must populate the MECID field in the request with a valid value. **When PAS is Non-secure** — The MECID value must be zero.
Host snoop to a cached HCM location or cached DCM location of a peer	False	The host snoop to a HCM location cached in a device must be processed normally. For a response to a stashing snoop where a DataPull is requested, the device must provide a DBID value.
	True	If the snoop response includes data, it must include: – The MECID value, if DataPull is 0. The inclusion of the MECID value in the snoop response enables the host to encrypt the data before writing back to HCM. – The DBID value, if DataPull is 1.

The permitted values and relationship between various fields is shown in Table B10.3.

The following key is used:

Y Yes, permitted.

- Not permitted.

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Table B10.3: Field value relationships in a device request to the host

SecSID1	Security state	PAS[2:0]	PAS description	Permitted combination
0b0	Non-Secure	0b000	Secure	-
		0b001	Non-secure	Y
		0b010	Root	-
		0b011	Realm	-
		0b100	System Agent	-
		0b101	Non-secure Protected	-
0b1	Realm	0b000	Secure	-
		0b001	Non-secure	Y
		0b010	Root	-
		0b011	Realm	Y
		0b100	System Agent	-
		0b101	Non-secure Protected	-

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